

Duncan T. Tulimieri, PhD

SENIOR QUANTITATIVE DEVELOPER

Chicago, IL

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Experience

New Age Alpha

New York | Remote

SENIOR QUANTITATIVE DEVELOPER

February 2023 - Present

- Built flexible and fast custom back testing software via microservices and scaled with Kubernetes reducing run time by 75%
- Built internal tools to detail portfolio point-in-time and time series metrics to expedite daily monitoring and research
- Built data pipelines in Azure Databricks to take disparate externally provided data and turn into cohesive tables for flexible and easy use
- Lead research projects on investigating and improving factor efficacy
- Backed all projects with test suites (unit and integration) to ensure easy bug-fix and refactoring and strict pre-commit hooks

ReproRehab [NIH-funded R25]

University of Southern California |

Remote

LEAD TEACHING ASSISTANT - DATA SCIENCE

October 2022 - Present

- Improve student experience through guiding other teaching assistants on best practices
- Upskill rehabilitation researchers data science techniques through weekly meetings and project assistance with Python, MATLAB, and git
- Encourage 1-on-1 help by hosting weekly office hours for troubleshooting Python, MATLAB, and git

KINARM

BKIN Technologies | Remote

TEACHING ASSISTANT - ROBOTICS

May 2020 - May 2022

- Educated neuroscience researchers and promoted research autonomy by teaching robotic programming skills (MATLAB, Simulink, Stateflow)

Education

University of Delaware

Newark, DE

DOCTORATE IN NEUROMECHANICS

June 2019 - May 2024

- Course work included, but not limited to: machine learning, neuromechanics, computational neuroscience, statistics, and data science

Denison University

Granville, OH

HEALTH, EXERCISE, AND SPORTS STUDIES & BIOLOGY

August 2015 - May 2019

- Department Fellow, Undergraduate Researcher, Tutor and Teaching Assistant, Strength and Conditioning Intern (Prentiss Hockey Performance)

Research Experience

University of Delaware

Newark, DE

DOCTORAL STUDENT

August 2019 - May 2024

- Designed and implemented end-to-end experiment design, pilot testing, data collection, and processing pipelines for 5 large-scale experiments
- Programmed 4 robotic tasks for KINARM Exoskeleton (using MATLAB and Simulink), 75% of which are used in continuing research
- Analyzed robotic experimental data by writing custom analysis scripts in MATLAB and Python
- Mentored 3 doctoral students and 3 undergraduate students to maximize learned content and conceptual/theoretical understanding
- Championed lab's main processing scripts (e.g., RedCap API), implementation of version control, and algorithm efficiency
- Collaborate with therapists (physical and occupational) and engineers to optimize projects

Denison University

Granville, OH

UNDERGRADUATE STUDENT

August 2015 - May 2019

- Designed, deployed, and analyzed survey using Qualtrics resulting in a publication to support Athletic Trainers' scope of practice expansion

Highlighted Projects

2024	Assessing and Training of Proprioception for Individuals With Chronic Stroke
2021	Development of a maze generation algorithm to be used as sensorimotor assessment
2022	Prediction of forest cover type on open-source data set with machine learning models
2019	Assessing the effect of speed and distance on kinesthetic matching in individuals with chronic stroke
2023	The perception of speed in the arms via psychophysical methods for individuals with chronic stroke
2020	Position matching with arms and eye movements for individuals with chronic stroke
2023	Robust optimization of minimum number of trials needed for experimental protocols
2022	Proprioceptive training for individuals with chronic stroke via robotic-joystick integration

Technical Skills

Languages (Python, MATLAB, SQL), Cloud & Infrastructure (Azure Databricks, Azure Data Factory, Kubernetes, git), Methodologies (Test Driven Development, Statistical Modeling, Machine Learning), Data Pipelines (Azure Databricks Jobs, Azure Data Factory), Automated Testing (Azure Pipelines)